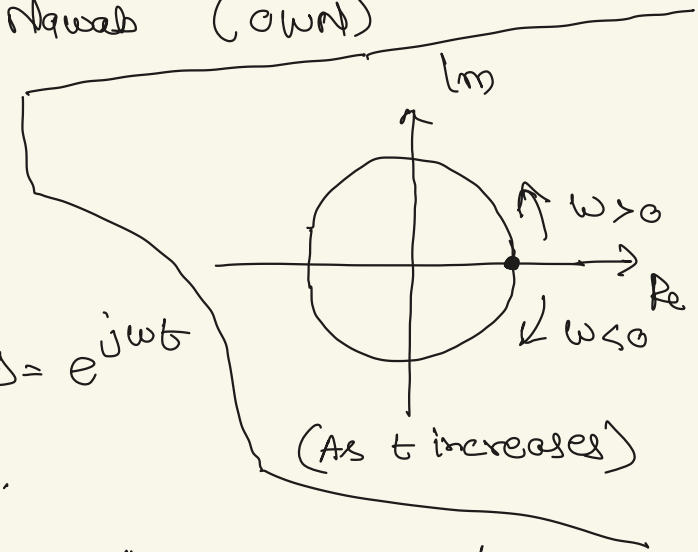
Lecture 3

I. Complex sinusoids

Reference: Chap 1. of "Signals & Systems"
By Oppenheim, Willsky, Nawab (OWN)
Second Edition.

Continuous-time signals only.



Defn: A signal of the form $x(t) = e^{j\omega t}$ is called a complex sinusoid.

Here $\omega \in \mathbb{R}$ is called "frequency" of the complex sinusoid, and its unit is rad/s.

How do we plot $x(t)$?

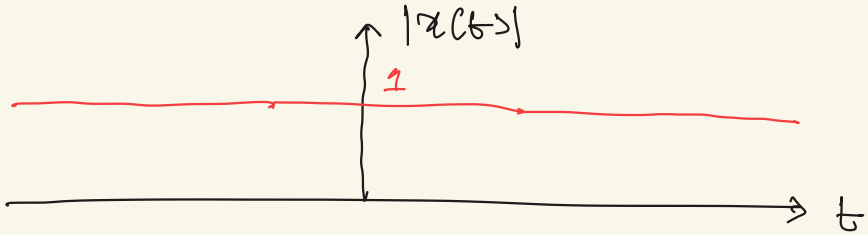
1. Plot $\mathrm{Re}(x(t))$, $\mathrm{Im}(x(t))$ vs. t

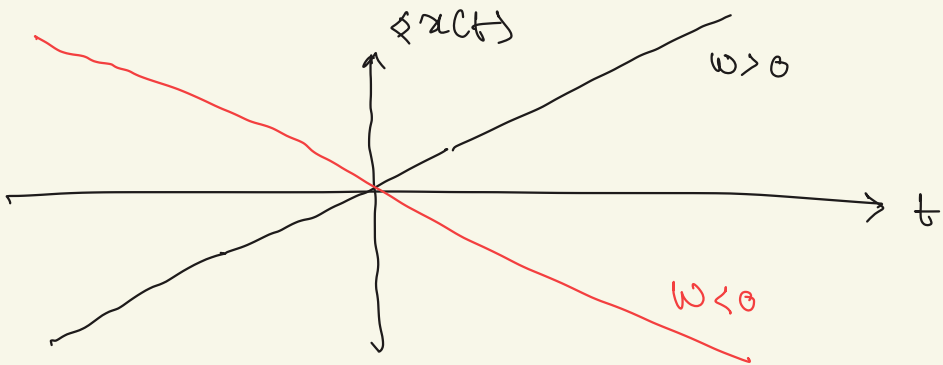
2. Plot $|x(t)|$ and $\measuredangle x(t)$ vs. t.

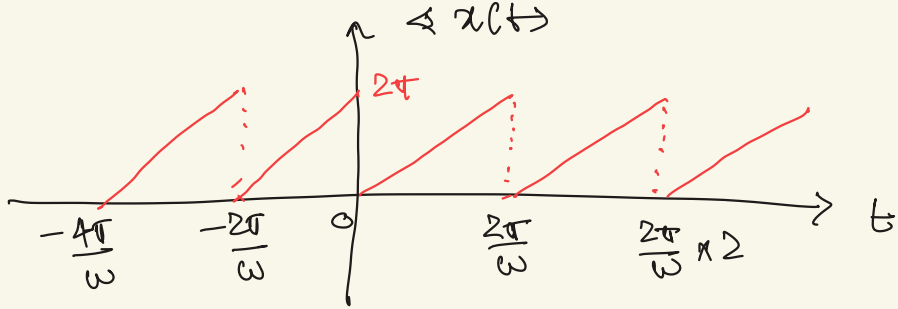
$e^{j\omega t}$

$|x(t)| = 1$

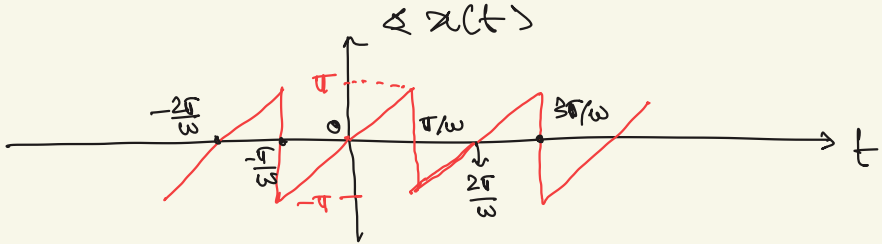
$\measuredangle x(t) = \omega t$







Both are OK.



<u>Remark:</u> Alternative definition/notation:

In some textbooks: $x(t) = e^{j2\pi f t}$

$f \rightarrow$ frequency, in Hz.

Relation between f and ω: $\omega = 2\pi f$

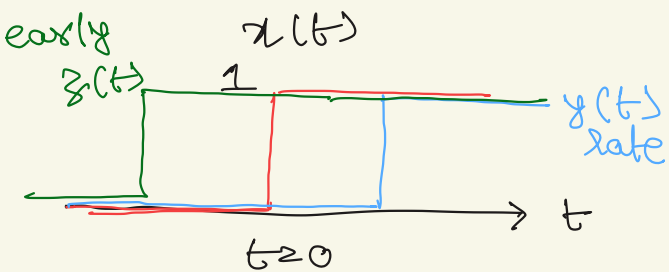
$d\omega = 2\pi\, df$

II. Transformations of the independent variable t

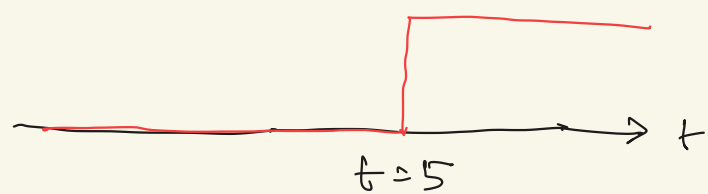
1. Time shifting

Given a signal $x(t)$ and a constant t_0, define the time-shifted signal:

$$y(t) = x(t - t_0) \quad \forall t$$



$t_0 = 5$ $\quad y(t) = x(t-5)$



$$x(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

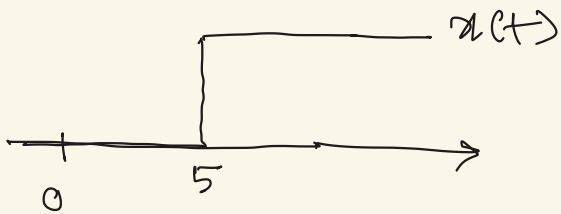
$z(t) = x(t+5)$

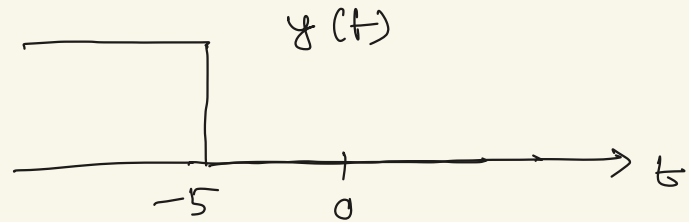
$x(t - 7.5) \rightarrow$ late

$x(t + 11/3) \rightarrow$ early.

2. <u>Time-reversal</u>

$$y(t) = x(-t) \quad \forall t$$

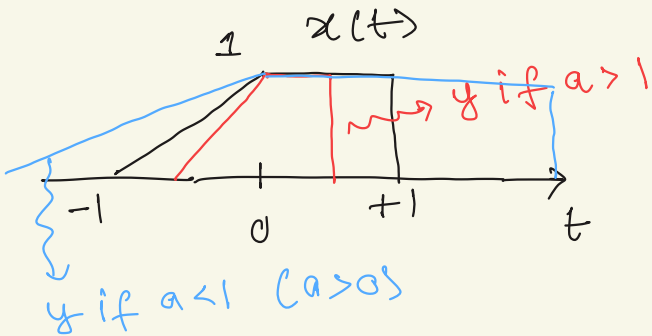


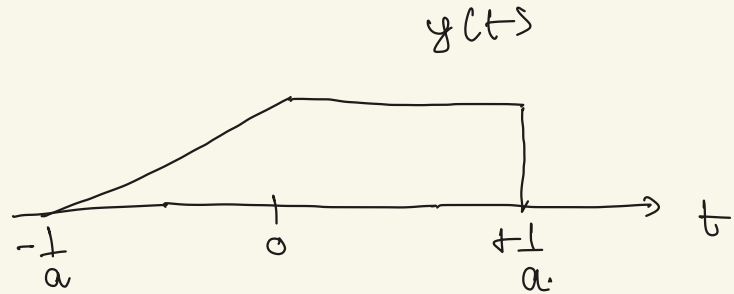


3. Time Scaling.

Choose any scaling factor $a > 0$.

Define $y(t) = x(at) \quad \forall t$





Blue: time expansion $\rightarrow a < 1$

Red: time compression $\rightarrow a > 1$

Transformations of t

Time shift : $t \rightarrow (t - t_0)$

Time reversal : $t \rightarrow (-t)$

Time scaling : $t \rightarrow (at)$

$\left\{ y(t) = x(t - t_0) \right.$

Ex: (1) $x \rightarrow$ timescale $a = 4 \rightarrow$ time shift $t_0 = 1$

$x(t) \rightarrow y(t) = x(4t) \rightarrow z(t) = y(t-1)$

$z(t) =$ time shift of $x(4t)$

$= x(4t)$ with t replaced by $t-1$

$= x(4(t-1))$

$= x(4t - 4)$

(2) $x \rightarrow$ time-shift $\rightarrow$ time-scale
$t_0 = 1$ with $a = 4$

$$x(t) \rightarrow y(t) = x(t-1) \rightarrow z(t) = y(4t)$$

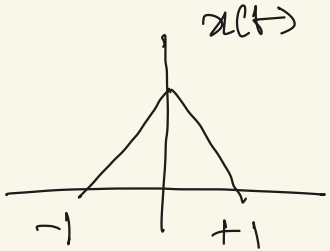
$\Rightarrow z(t)$ is time scaling of $x(t-1)$

$= x(t-1)$ with t replaced by $4t$

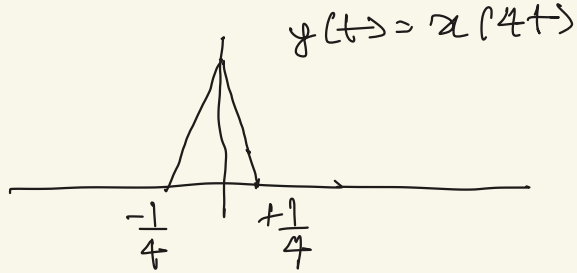
$= x(4t-1)$

Warning: These operations are non-commutative: changing their order of applications can change the final outcome.

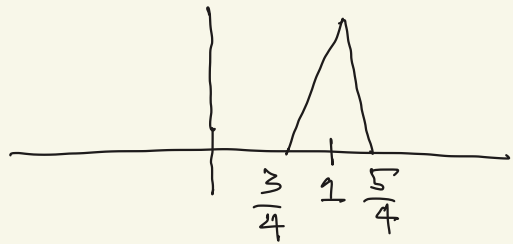
(1)



$a=4$



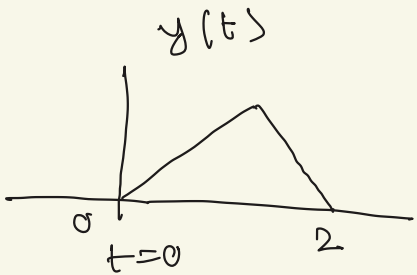
$z(t) = y(t-1)$



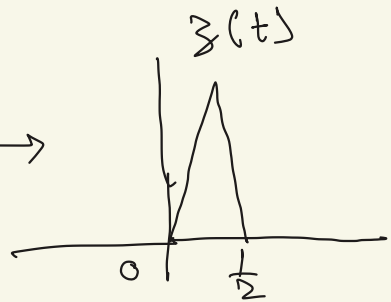
$x(4t-4)$

$t_0 = 1$

(2)



$a=4$



$x(4t-1)$